ARJUNA JEE 2027

Lecture -04 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A Geometric Progression (G.P.)

B Sum of G.P.

C Different types of G.P.

D Properties of G.P.



Question Practice

Question [JEE Mains-2024]

For $A = \{n \in [100, 700] \cap \mathbb{N}, n$ is neither a multiple of 3 nor a multiple of 4$\}$. Then the number of elements in A is

82%

A 290

B 280

C 300

D 310

$\{100, 101, 102, \ldots . 400\}$

Total elem. $= 400 - 100 + 1$

$= 601.$

Question [JEE Mains-2022] *Sawal puchne ka tarika thoda casual hai.*

The sum of all the elements of the set

$\{\alpha \in \{1, 2, \ldots\ldots, 100\} : \text{HCF}\,(\alpha, 24) = 1\}$ is _____.

[Ans. : (1633)]

Question [JEE Mains-2021]

[Ans. : (1000)]

The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is

$\{1000, 1001, \ldots\ldots\ldots 9999\}$ → Total = 9000.

GCD OR HCF $(??, 18) = 3^1$

$??$ → $2^0\, 3^1 5^? 7^?$; 18 → $2^1 \cdot 3^2$

Those numbers whose P.f. contains $2^0 \cdot 3^1$

OR

Odd & multiple of 3 but not a multiple of 9

Question [JEE Mains-2021]

[Ans. : (1000)]

The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is

$\{1000, 1001, \ldots\ldots, 9999\}$

$n(f) = 1500 - 500$

HCF or GCD $(__, 18) = 3$

* odd + multiple of 3 ✓ + multiple of 9 ✗

odd + multiple of 3 :

A.P.

$\{1005, 1011, \ldots\ldots 9999\}$ → no. of Terms = 1500 ✓

+6 +6

odd + multiple of 9 :

A.P.

$\{1017, 1035 \ldots\ldots, 9999\}$ → no. of Terms = 500

+18 +18

Question [JEE Mains-2023]

2, 7, 12, 17, 22, 27, 32, 37, 42, 47 . . . .

[Ans. : (321)]

The sum of the common terms of the following three arithmetic progressions.

$3, 7, 11, 15, \ldots\ldots\ldots\ldots, 399$

$2, 5, 8, 11, \ldots\ldots\ldots\ldots\ldots\ldots 359$ and

$2, 7, 12, 17, \ldots\ldots, 197$, is equal to

$11, 23, 35, 47, \ldots\ldots\ldots 348$

CT's (A.P.) → 11, 23, 35, 47, ... 359 (30 terms)

d = 12 (+12 +12 +12)

d = 5, d = 12 ⟩ LCM(5, 12) = 60

CT's: {47, 107, 167} (+60, +60)

Sum = 167 + 107 + 47 = 321 Ans.

Question [JEE (Adv.)-2018, Paper-1]

[Ans. : (3748)]

Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, ... Then, the number of elements in the set $X \cup Y$ is _____.

$X = \{1, 6, 11, 16, \ldots \text{upto 2018 Terms}\}$ (+5, +5) $\rightarrow d_1 = 5$, $T_{2018} = 1 + 2017 \times 5$

$Y = \{9, 16, 23, \ldots \text{upto 2018 Terms}\}$ (+7, +7) $\rightarrow d_2 = 7$, $T_{2018} =$

$$n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$$

$$= 2018 + 2018 - 288$$

$$= 3748 \text{ Ans.}$$

$X \cap Y = \{16, 51, \ldots T_n\}$ (+35)

$d = \text{LCM}(5, 7) = 35$

$T_n = 16 + (n-1)35$

$= 35n - 19 \leq 1 + 2017 \times 5$

$n_{\text{Largest}} = 288$

Challenger Question [JEE Mains-2023]

Calculative

[Ans. : (910)]

Let a_1 (+ve), $a_2, a_3, \ldots\ldots$ be in an arithmetic progression of positive terms.

Let $A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots + a_{2k-1}^2 - a_{2k}^2$.

If $A_3 = -153, A_5 = -435$ and $a_1^2 + a_2^2 + a_3^2 = 66$, then $a_{17} - A_7$ is equal to

$a_1^2 + a_2^2 + a_3^2 = 66$ → No use.

$$A_k = (a_1+a_2)\underbrace{(a_1-a_2)}_{-d} + (a_3+a_4)\underbrace{(a_3-a_4)}_{-d} + \ldots\ldots + (a_{2k-1}+a_{2k})\underbrace{(a_{2k-1}-a_{2k})}_{-d}$$

$$= -d\{a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + \ldots\ldots + a_{2k-1} + a_{2k}\}$$

A.P.

$a = a_1,\ d,\ n = 2k$

$$A_k = -d\left\{\frac{2k}{2}\left(2a_1 + (2k-1)d\right)\right\}$$

$A_7 = -861$

$a_{17} = a_1 + 16d$

$= 1 + 16 \times 3$

$a_{17} = 49$

$49 - (-861)$

910

fnf.

$$A_K = -\frac{d}{2}\left\{\frac{2K}{2}\left(2a_1+(2K-1)d\right)\right\}$$

$$A_K = -3K\left(2+(2K-1)3\right)$$

$$A_7 = -21\left(2+13\times 3\right) = -21\times 41$$

$A_3 = -153$

$A_5 = -435$

$$\begin{cases} a_1 = 1 \\ d = 3 \end{cases}$$

$$A_K = -\frac{dK}{2}\left(2a_1+(2K-1)d\right)$$

$K=3$: $A_3 = +\frac{d}{2}(3)\left(2a_1+5d\right) = +153$

$K=5$: $A_5 = +\frac{d}{2}(5)\left(2a_1+9d\right) = +435$

$$\frac{6a_1+15d}{5\left(2a_1+9d\right)} = \frac{51}{29}$$

$$6\times 29a_1 + 15\times 29d = 102a_1 + 9\times 51d$$

$$1740a_1 - 102a_1 = 459d - 435d$$

$$3\times 78a_1 = 24d$$

$$3a_1 = d \quad \checkmark$$

$$A_3 = -\frac{d}{\downarrow}(3)\left(2a_1+5\frac{d}{\downarrow}\right) = -153$$

$$+(3a_1)\times 3\left(2a_1+5\times 3a_1\right) = +153$$

$$17a_1^2 = 17 \rightarrow a_1^2 = 1 \rightarrow a_1 = \pm 1 \quad \checkmark$$

Question [JEE Mains-2023]

Calculative + Tough

For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and $r = pq + 1$ such that $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in A.P. with common difference 1/2. Then $r - p - q$ is equal to

(A) 2

(B) 6

(C) 12

(D) −6

Hint

$x^{pq^2} = y^{qr} = z^{p^2r} = t$ (say)

$x = (t)^{\frac{1}{pq^2}}$ $y = (t)^{\frac{1}{qr}}$ $z = (t)^{\frac{1}{p^2r}}$

Sol.

Question

$2b = a + c$

If a, b, c be in A.P. prove that

(i) $a^2(b + c), b^2(c + a), c^2(a + b)$ are in A.P.

(ii) $\frac{ab+ac}{bc}, \frac{bc+ba}{ca}, \frac{ca+bc}{ab}$ are in A.P.

Say:

(i) $a^2(b+c)$, $b^2(c+a)$, $c^2(a+b)$ are in A.P.

$$b^2(c+a) - a^2(b+c) = c^2(a+b) - b^2(c+a)$$

$$b^2c + b^2a - a^2b - a^2c = c^2a + c^2b - b^2c - b^2a$$

$$c(b^2-a^2) + ab(b-a) = bc(b-c) + a(c^2-b^2)$$



$a, b, c \rightarrow$ A.P.

1 2 3

-1, 0, 1

Say:

(i) $a^2(b+c)$, $b^2(c+a)$, $c^2(a+b)$ are in A.P.

$$b^2(c+a) - a^2(b+c) = c^2(a+b) - b^2(c+a)$$

$$b^2c + b^2a - a^2b - a^2c = c^2a + c^2b - b^2c - b^2a$$

$$c(b^2-a^2) + ab(b-a) = bc(c-b) + a(c^2-b^2)$$

$(b-a)(b+a)$ $\qquad$ $(c-b)(c+b)$

$$(b-a)\{c(b+a)+ab\} = (c-b)\{bc+a(c+b)\}$$

$(cb+ac+ab)$ $\qquad$ $(bc+ac+ab)$

$$b-a = c-b$$

$$2b = a+c \rightarrow \text{Hence proved.}$$

dengfay!!

Question

$2b = a + c$.

If a, b, c be in A.P. prove that

(i) $a^2(b + c), b^2(c + a), c^2(a + b)$ are in A.P.

(ii) $\frac{ab+ac}{bc}, \frac{bc+ba}{ca}, \frac{ca+bc}{ab}$ are in A.P. ✓

Objective.

Youtube Method.

value putting

$a, b, c \rightarrow$ A.P.

1 2 3

-1, 0, 1

Say:

(i) $a^2(b+c)$, $b^2(c+a)$, $c^2(a+b)$ are in A.P.

using Prop.

$\div abc$

$\frac{a^2(b+c)}{abc}, \frac{b^2(c+a)}{abc}, \frac{c^2(a+b)}{abc} \rightarrow$ A.P.

$\frac{ab+ac}{bc}, \frac{bc+ba}{ac}, \frac{ca+cb}{ab} \rightarrow$ A.P.

(i) $a^2(b+c)$, $b^2(c+a)$, $c^2(a+b)$ are in A.P.

using Prop. ★★★★★★★

$\div abc$

$\frac{a^2(b+c)}{abc}$, $\frac{b^2(c+a)}{abc}$, $\frac{c^2(a+b)}{abc}$ → A.P.

$\frac{ab+ac}{bc}$, $\frac{bc+ba}{ac}$, $\frac{ca+cb}{ab}$ → A.P.

$1+\frac{ab+ac}{bc}$, $1+\frac{bc+ba}{ac}$, $1+\frac{ca+cb}{ab}$ → A.P.

$\div(ab+bc+ca)$

$\frac{ab+bc+ca}{bc}$, $\frac{ab+bc+ca}{ac}$, $\frac{ab+bc+ca}{ab}$ → A.P.

$\frac{1}{bc}$, $\frac{1}{ac}$, $\frac{1}{ab}$ → A.P.

$\times abc$

$\frac{abc}{bc}$, $\frac{abc}{ac}$, $\frac{abc}{ab}$ → A.P.

a, b, c → A.P.

Hence Proved

GEOMETRIC PROGRESSION

$\times r$ $\times r$ $\times r$

$a, ar, ar^2, ar^3, \ldots\ldots\ldots\ldots \rightarrow$ G.P.

(i) First term = a.

(ii) Common ratio = r. $r = \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \ldots\ldots$

(iii) n^{th} term (T_n) = $a(r)^{n-1}$.

Question [JEE Mains-2024]

Calculative

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th term is

70%

A 96

B 78

C 91

D 84

$T_2 + T_6 = \frac{70}{3}$

$T_3 \times T_5 = 49$

G.P. $\rightarrow a, r$.

$ar^1 + ar^5 = \frac{70}{3}$ —(1)

$ar^2 \times ar^4 = 49$ —(2)

$a = \frac{7}{3\sqrt{3}}$, $r = \sqrt{3}$.

$T_4 + T_6 + T_8 = ??$

$ar^3 + ar^5 + ar^7$

$ar^3(1 + r^2 + r^4)$

$\frac{7}{3\sqrt{3}} \times 3\sqrt{3}(1+3+9) = 7 \times 13 = 91$

$ar^1 + ar^5 = \frac{70}{3}$ —(1)

$ar^2 \times ar^4 = 49$ —(2)

$a^2r^6 = 7^2$

$(ar^3)^2 = 7^2$

$ar^3 = \pm 7$ ✓

T_4 ⊖ ×

$ar^3 = 7$

$a = \frac{7}{r^3}$

$ar(1 + r^4) = \frac{70}{3}$

$\frac{r}{r^{3+2}} \times r\left(1 + r^4\right) = \frac{70}{3}$ (70 → 10)

$\frac{1 + r^4}{r^2} = \frac{10}{3}$

$r^2 = \frac{1}{3}$ or 3

$r = \pm\frac{1}{\sqrt{3}}, \pm\sqrt{3}$

$3r^4 + 3 = 10r^2$

$3r^4 - 10r^2 + 3 = 0$

$3r^4 - 9r^2 - r^2 + 3 = 0$

$3r^2(r^2 - 3) - 1(r^2 - 3) = 0$

$(3r^2 - 1)(r^2 - 3) = 0$

$r^2 = \frac{1}{3}$ or 3.

$ar^3 = 7$ → $a = \frac{7}{r^3}$

$r^2 = \frac{1}{3}$ or 3 → $r = \pm\frac{1}{\sqrt{3}}, \pm\sqrt{3}$

$r = \frac{1}{\sqrt{3}}$ ✗

$a = \frac{7}{\left(\frac{1}{\sqrt{3}}\right)^3} = 7 \times 3\sqrt{3}$

G.P. → $a, ar, ar^2, ar^3 \ldots\ldots$

$7 \times 3\sqrt{3}, \quad 7 \times 3\sqrt{3} \times \frac{1}{\sqrt{3}}, \quad 7 \times 3\sqrt{3} \times \frac{1}{3}, \ldots\ldots$

$21\sqrt{3}, \quad 21, \quad 7\sqrt{3}, \ldots\ldots$ → decreasing G.P.

$r = \sqrt{3} \longrightarrow a = \frac{7}{(\sqrt{3})^3} = \frac{7}{3\sqrt{3}}$

G.P. → $\frac{7}{3\sqrt{3}}, \frac{7}{3}, \frac{7}{\sqrt{3}}, 7, 7\sqrt{3} \ldots$ → increasing G.P.

If (r) → (-ve)

Terms of G.P. are with alternate +/− sign.





SUM OF n-TERMS OF GEOMETRIC PROGRESSION

$S_n = a + ar + ar^2 + ar^3 + \ldots\ldots\ldots + ar^{n-2} + ar^{n-1}$

multiply by r

$rS_n = 0 + ar + ar^2 + ar^3 + \ldots\ldots\ldots + ar^{n-1} + ar^n$

$\ominus$

$S_n - rS_n = a - ar^n$

$S_n(1-r) = a - ar^n$

$$S_n = a\left(\frac{1-r^n}{1-r}\right) = a\left(\frac{r^n-1}{r-1}\right)$$

$$*** \quad \boxed{S_n = a\left(\frac{1-r^n}{1-r}\right) = a\left(\frac{r^n-1}{r-1}\right)}$$

, $(r \neq 1)$, $n \geq 1$, $n \in N$

$\left\{\begin{array}{l} a = \text{first Term} \\ r = \text{C.R.} \\ n = \text{total no. of Terms} \end{array}\right\}$

If $(r=1)$:

$$S = a + ar + ar^2 + \ldots\ldots + ar^{n-1}$$

$$S = a + a + a + \ldots\ldots + a$$

$\left(\underset{d=0}{\text{A.P.}}\checkmark , \underset{r=1}{\text{G.P.}}\checkmark\right)$

$$\boxed{S_n = na}$$

• RFR

$S_n = \dfrac{a(r^n-1)}{(r-1)}$

Ques:-

(i) $a = 3,\ r = \frac{1}{2},\ n = 5$ $\longrightarrow$ $S_5 = 3 \times \dfrac{\left(\left(\frac{1}{2}\right)^5 - 1\right)}{\left(\frac{1}{2} - 1\right)} = \dfrac{3\left(1 - 2^5\right)}{2^5\left(\frac{-1}{2}\right)}$

$= \dfrac{3(2^5-1)}{2^4}$

(ii) $a = 2,\ T_6 = 32$

$ar^5 = 32$

$2r^5 = 32$

$r^5 = 16 = 2^4$

$r = \left(2^4\right)^{\frac{1}{5}} = 2^{\frac{4}{5}}$

$S_{11} = a\left(\dfrac{r^n - 1}{r-1}\right)$

$= 2\left(\dfrac{\left(2^{\frac{4}{5}}\right)^{11} - 1}{2^{\frac{4}{5}} - 1}\right)$

Question [JEE Mains-2020]

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is:

[JEE Mains-2020]

48%

A $\frac{2}{13}(3^{50}-1)$

B $\frac{1}{26}(3^{50}-1)$

C $\frac{1}{26}(3^{49}-1)$

D $\frac{1}{13}(3^{50}-1)$

$T_2 + T_3 + T_4 = 3$.

$T_6 + T_7 + T_8 = 243$.

$ar + ar^2 + ar^3 = 3$

$ar^5 + ar^6 + ar^7 = 243$.

ar^4

$ar^4(1 + r + r^2) = 243$.

$ar(1 + r + r^2) = 3$.

$\Rightarrow r^4 = 81 = 3^4$

$r = \pm 3$.

$r = 3$.

$ar(1 + r + r^2) = 3$.

$3a(13) = 3$.

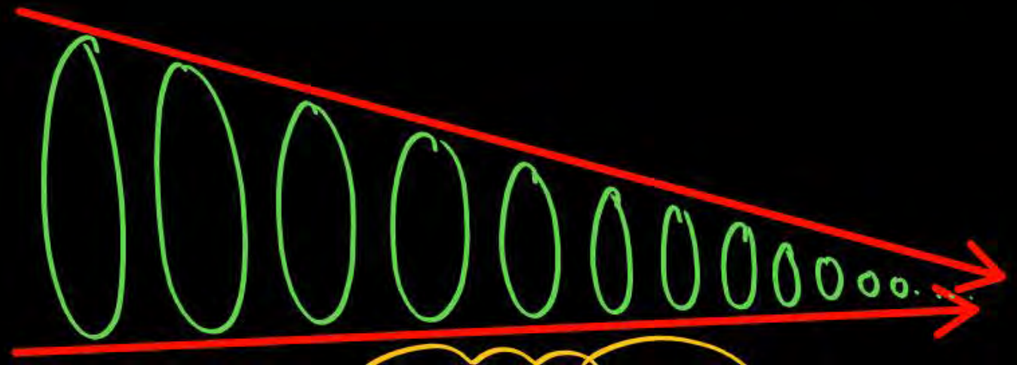
$a = \frac{1}{13}$

$S_{50} = a\left(\frac{r^{50}-1}{r-1}\right)$

$= \frac{1}{13}\left(\frac{3^{50}-1}{3-1}\right)$

$= \frac{1}{26}(3^{50}-1)$.

SUM OF INFINITE-TERMS OF GEOMETRIC PROGRESSION



Converging Series

S_∞ exists

$-1 < r < 1$, $(r \neq 0)$

$$S_n = a\left(\frac{1-r^n}{1-r}\right)$$

$\left\{\begin{matrix} -1<r<1 \\ n \to \infty \end{matrix}\right\}$ $r^n \to 0$

$$S_\infty = \frac{a}{1-r}$$

If $-1 < r < 1$

\# then for $n \to \infty \Rightarrow r^n \to 0$.

\# $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \ldots\ldots\infty = \frac{a}{1-r} = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{\frac{1}{3}}{2/3} = \frac{1}{2}$ ✓

$a = \frac{1}{3}, r = \frac{1}{3}$

Question [JEE Mains-2024]

An arithmetic progression is written in the following way

2

5 8

11 14 17

20 23 26 29

The sum of all the terms of the 10th row is _____.

Question [JEE Mains-2023]

Ans: (8)

Let $a_1, a_2, \ldots\ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then $12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}} + \ldots + \frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$ is equal to

(Hw)

Rationalise each term.

DO IT BY YOURSELF (DIBY)

DIBY-01

Let a_n be the nth term of an $A.P.$ If $S_n = a_1 + a_2 + a_3 + \cdots . + a_n = 700, a_6 = 7$ and $S_7 = 7$, then a_n is equal to: [JEE Mains-2025]

(A) 56 (B) 65 (C) 64 (D) 70

[Ans. : (C)]

DIBY-02

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to [JEE Mains-2025]

(A) –120 (B) –1200 (C) –1020 (D) –1080

[Ans. : (D)]

DIBY-03

The roots of the quadratic $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to [JEE Mains-2025]

[Ans. : 474]

DIBY-04

In an arithmetic progression, if $S_{40} = 1030$ and $S_{12} = 57$, then $S_{30} - S_{10}$ is equal to

[JEE Mains-2025]

(A) 505 (B) 525 (C) 510 (D) 515

[Ans. : (D)]

DIBY-05

If $a_1, a_2, a_3, \ldots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is

[JEE Mains-2019]

(A) 200 (B) 280 (C) 120 (D) 260

[Ans. : (A)]

DIBY-06

If 19th terms of non-zero A.P. is zero, then its $(49^{th}$ term$) : (29^{th}$ term$)$ is:

[JEE Mains-2019]

(A) 4 : 1 (B) 1 : 3 (C) 3 : 1 (D) 2 : 1

[Ans. : (C)]

DIBY-07

Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is:

[JEE Mains-2021]

(A) 2 (B) 6 (C) 8 (D) 4

[Ans. : (B)]

DIBY-08

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\frac{a_1+a_2+\cdots+a_{10}}{a_1+a_2+\cdots+a_p} = \frac{100}{p^2}, p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to:

[JEE Mains-2021]

(A) $\frac{21}{19}$ (B) $\frac{100}{121}$ (C) $\frac{121}{100}$ (D) $\frac{19}{21}$

[Ans. : (A)]

DIBY-09

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to

[JEE Mains-2021]

[Ans. : 3]

DIBY-10

If the 10th term of an A.P. is $\frac{1}{20}$ and its 20th term is $\frac{1}{10}$, then the sum of its first 200 terms is

[JEE Mains-2020]

(A) 100 (B) $50\frac{1}{4}$ (C) $100\frac{1}{2}$ (D) 50

[Ans. : (C)]

DIBY-11

If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \cdots$ upto n^{th} term is 488 and the nth term is negative, then

[JEE Mains-2020]

(A) $n = 41$

(B) n^{th} term is $-4\frac{2}{5}$

(C) $n = 60$

(D) n^{th} term is -4

[Ans. : (D)]

DIBY-12

Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $\frac{-1}{2}$, then the greatest number amongst them is

[JEE Mains-2020]

(a) $\frac{21}{2}$

(B) 16

(C) 27

(D) 7

[Ans. : (B)]

DIBY-13

If $3^{2\sin 2\alpha-1}, 14$ and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some α, then the sixth term of this A.P. is: [JEE Mains-2020]

(A) 65 (B) 78 (C) 81 (D) 66

[Ans. : (D)]

DIBY-14

If $a_1, a_2, a_3.$ and $b_1, b_2, b_3.$ are in A.P., and $a_1 = 2, a_{10} = 3,\ a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4 b_4$ is equal to. [JEE Mains-2022]

(A) $\frac{35}{27}$ (B) 1 (C) $\frac{27}{28}$ (D) $\frac{28}{27}$

[Ans. : (D)]

DIBY-15

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to: [JEE Mains-2024]

(A) 125 (B) 150 (C) 180 (D) 160

[Ans. : (B)]

DIBY-16

Consider two sets A and B, each containing three numbers in $A.P$. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $p - q$ is equal to [JEE Mains-2025]

(A) 600 (B) 450 (C) 630 (D) 540

[Ans. : (D)]

DIBY-17

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is: [JEE Mains-2025]

(A) 84 (B) 90 (C) 108 (D) 122

[Ans. : (B)]

DIBY-18

Number of 4 digit integers less than 2800 which are either divisible by 3 or by 11 is equal to ?

[JEE Mains-2023]

[Ans. : 710]

DIBY-19

If where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192, \sum_{i=1}^{n/2} a_{2i} = 120,$, then n is equal to [JEE Mains-2022]

(A) 48 (B) 96 (C) 92 (D) 104

[Ans. : (B)]

DIBY-20

Let S_1 be the sum of first $2n$ term of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to

[JEE Mains-2021]

(A) 5000 (B) 1000 (C) 7000 (D) 3000

[Ans. : (D)]

DIBY-21

The sum of all natural numbers 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is

[JEE Mains-2019]

(A) 3221 (B) 3121 (C) 3203 (D) 3303

[Ans. : (B)]

DIBY-22

The series of positive multiples of 3 is divided into sets : {3}, {6, 9, 12}, {15, 18, 21, 24, 27}, ... Then the sum of the elements in the 11th set is equal to

[JEE Mains-2022]

[Ans. : 6993]

DIBY-23

Let $a_1, a_2, \ldots a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{2i-1}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to

[JEE Mains-2019]

(A) 52 (B) 57 (C) 47 (D) 42

[Ans. : (A)]

DIBY-24

Let $a_1, a_2, \ldots a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \cdots + a_{2020}) + a_{2024} = 223$. Then $a_1 + a_2 + a_3 + \cdots + a_{2024}$ is equal to

[JEE Mains-2025]

[Ans. : 11132]

DIBY-25

If p^{th}, q^{th} and r^{th} term of an A.P. are respectively a, b and c, then the value of $a(q-r) + b(r-p) + c(p-q)$ is

(A) 0 (B) 1 (C) $p+q+r$ (D) abc

[Ans. : (A)]

DIBY-26

Find the value of the sum $S = 1002 - 992 + 982 - 972 + 962 - 952 + \cdots + 22 - 12$

(A) 4950 (B) 10100 (C) 5150 (D) 5050

[Ans. : (D)]

DIBY-27

The sum of first 3 terms of an A.P. is 27 and the sum of their squares is 293, then first term and common difference of A.P. is

(A) First term $= 3$, Common difference $= \pm 6$

(B) First term = 4, Common difference = 5

(C) First term = 4 or 14, Common difference = ± 5

(D) First term = 2 or 16, Common difference $= \pm 7$

[Ans. : (C)]

DIBY-28

The number of terms in an A.P. is even, the sum of the odd numbered terms is 24 & sum of the even numbered terms is 30 and the last term exceeds the first term by $10\frac{1}{2}$. Find the number of terms.

(A) 4 (B) 6 (C) 8 (D) 12

[Ans. : (C)]

It is never about end result

It's always about Journey

Thank You

#futureIITians